

Computer Networks

Lecture 4: OSI Model

Content

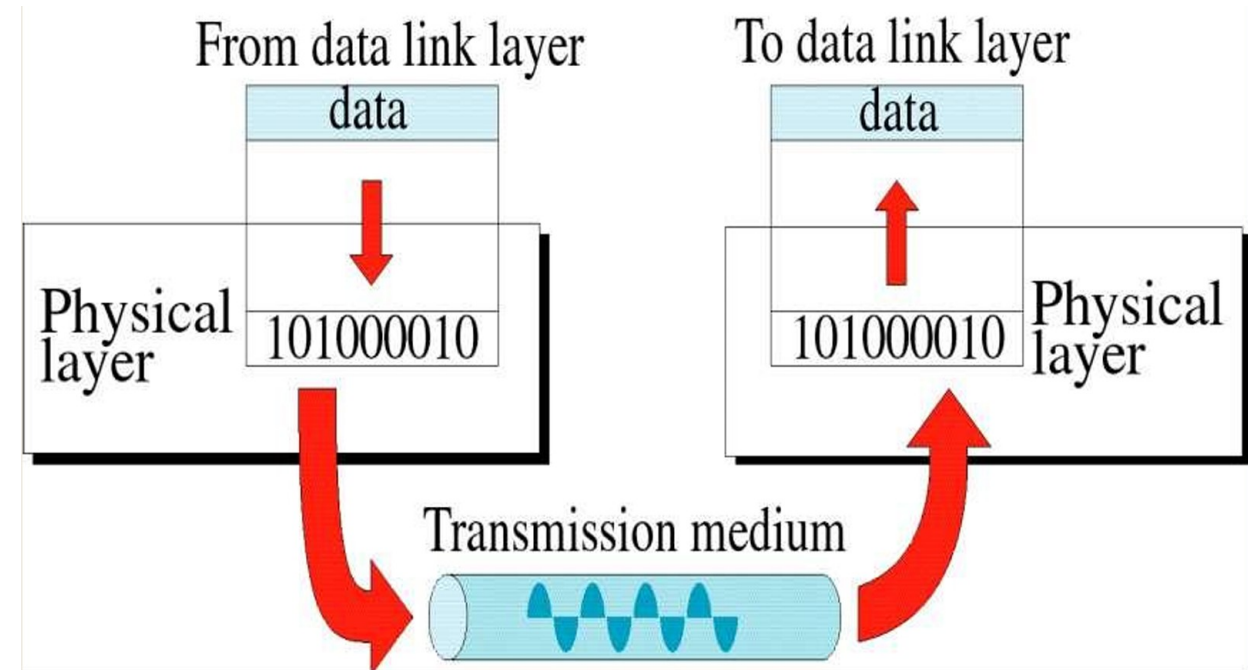


- **OSI Model**

Physical Layer

Physical layer is the bottom(layer 1) of OSI model.

- It is responsible for the actual physical connection between the devices.
- The physical layer is responsible for movements of individual bits from one node to next.



Functions of Physical Layer

- Physical Characteristics of medium: How devices are physically connected
- Representation of bits: How bits are represented and carried over the transmission medium.
- The transmission rate: the number of bits sent each second
- Synchronization of bits: the sender and the receiver clocks must be synchronized.
- Physical topology: How devices are connected.
- Transmission mode: The direction of transmission between two devices

Physical Layer

Physical Layer is a raw transmission facility

Can only send and receive bits (0s and 1s)

Is there any limitation ?

Physical layer has no idea:

- If bits are lost
- If bits are duplicated
- If bits are corrupted
- Where one message starts or ends It is like a wire that blindly carries signals

What we expect?

A reliable link where errors can be detected, sender and receiver are synchronized.

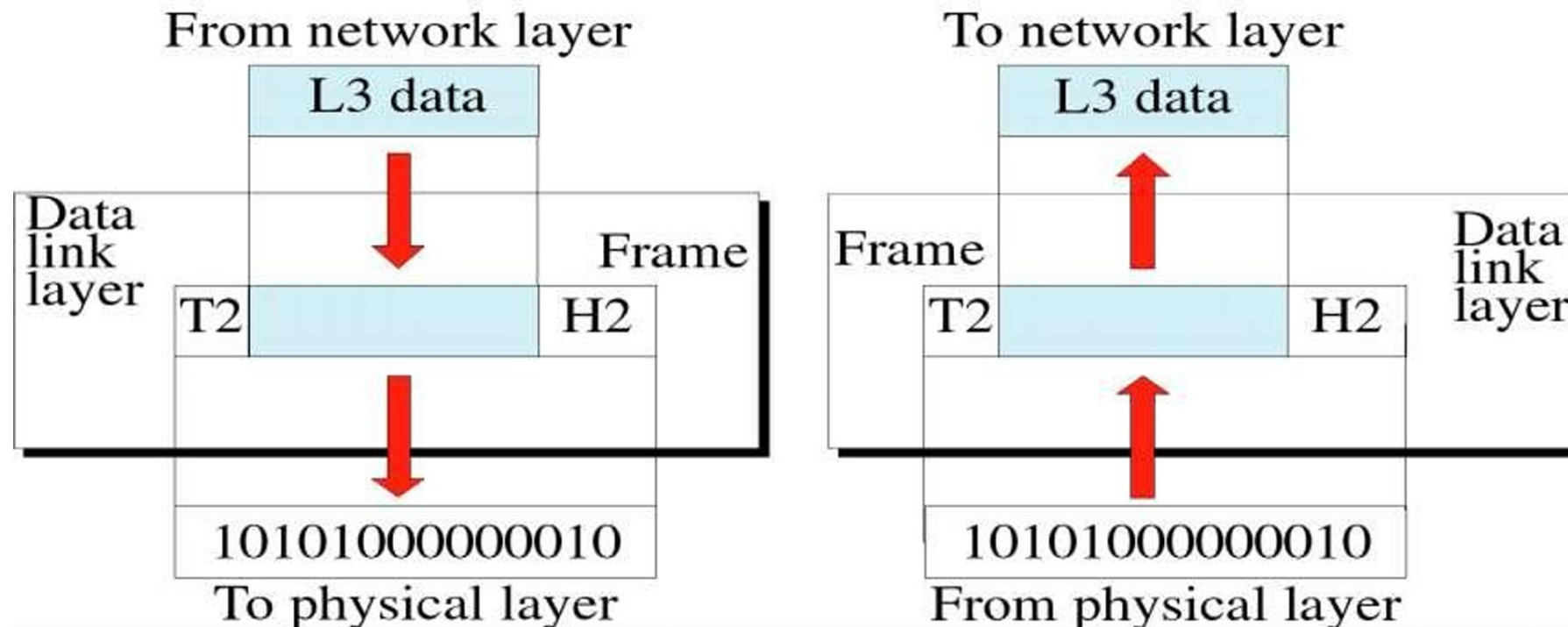
FRAME 1 → 010101010101...

FRAME 2 → 110010101011...

FRAME 3 → 001011010101...

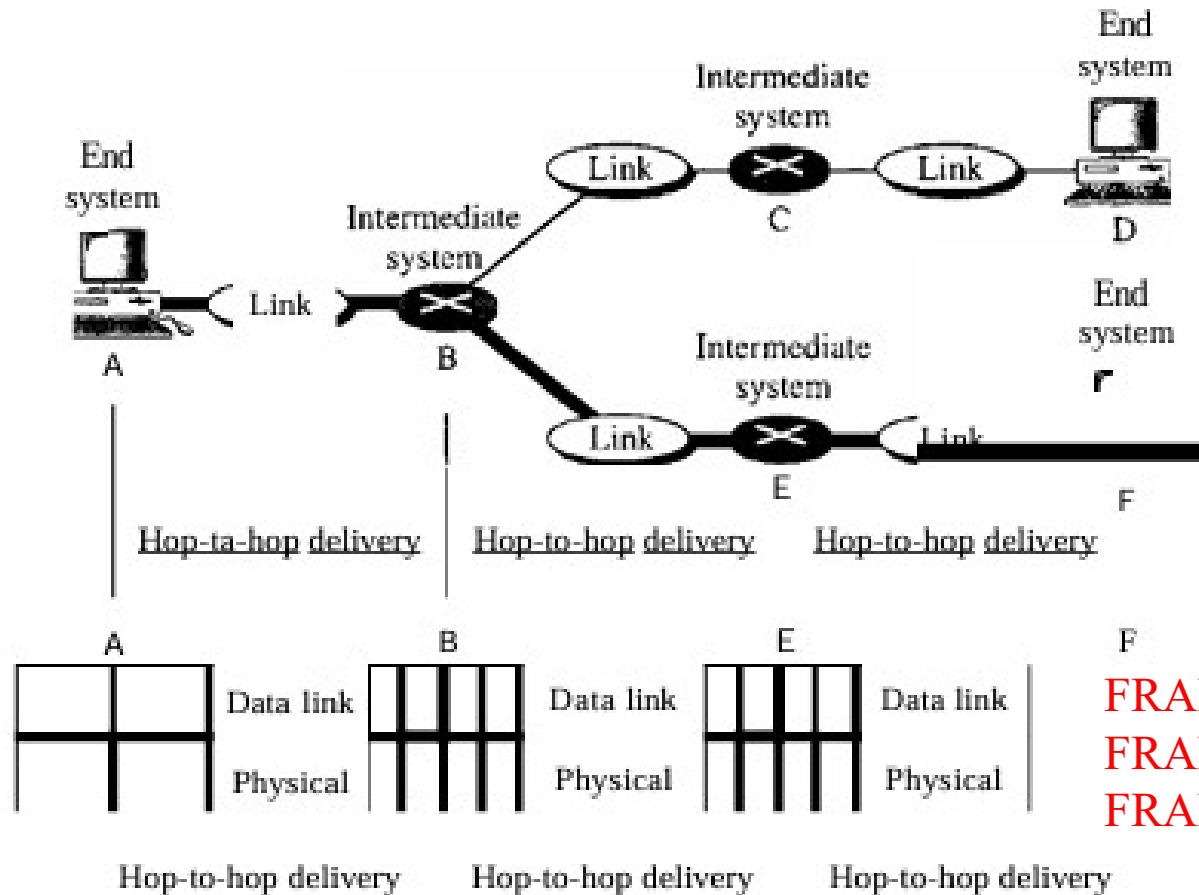
Data Link Layer

The Data Link layer is responsible for reliable node-to-node (hop-by-hop) delivery of data by transforming the physical layer's raw bit stream into error-controlled, well-defined frames.



Data Link Layer

Hop-to-hop(node-to-node) delivery by the data link layer.



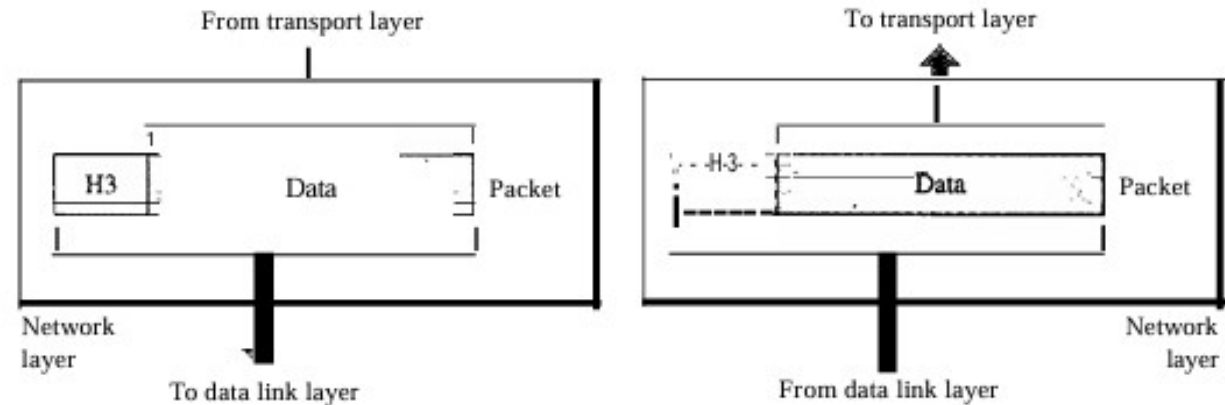
Functions of Data Link Layer

- Framing
- Physical addressing
- Flow control
- Error control
- Access control

Network Layer

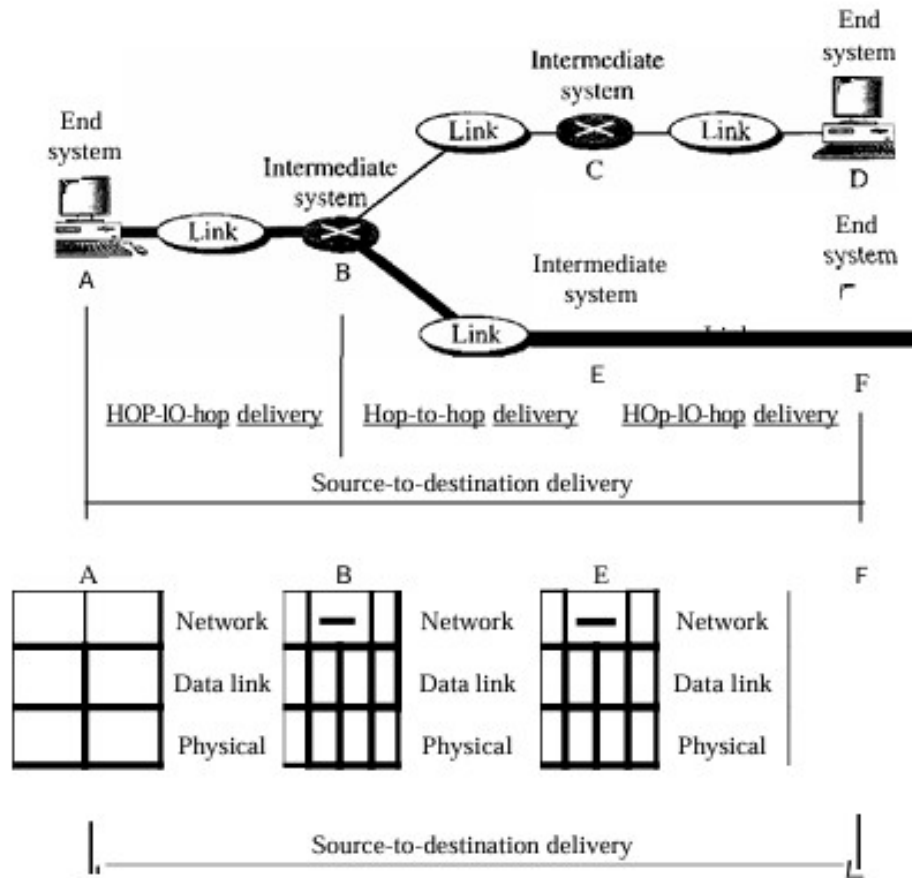
The network layer is responsible for the source-to-destination delivery of a packet, possibly across multiple networks (links).

The data link layer oversees the delivery of the packet between two systems on the same network (links), the network layer ensures that each packet gets from its point of origin to its final destination.



Network Layer

Source to destination delivery by the network layer.



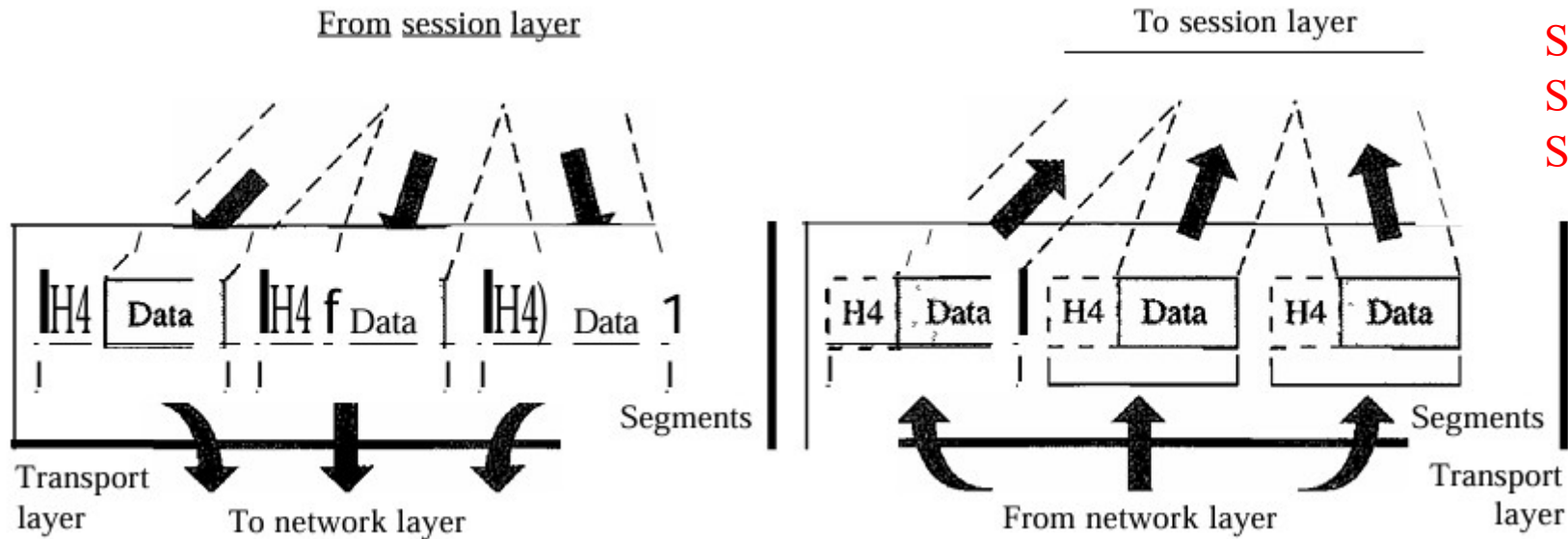
PACKET 1 : [Source IP | Destination IP | SEGMENT 1]
 PACKET 2 : [Source IP | Destination IP | SEGMENT 2]
 PACKET 3 : [Source IP | Destination IP | SEGMENT 3]

Functions of Network Layer

- Logical addressing
- Routing

Transport Layer

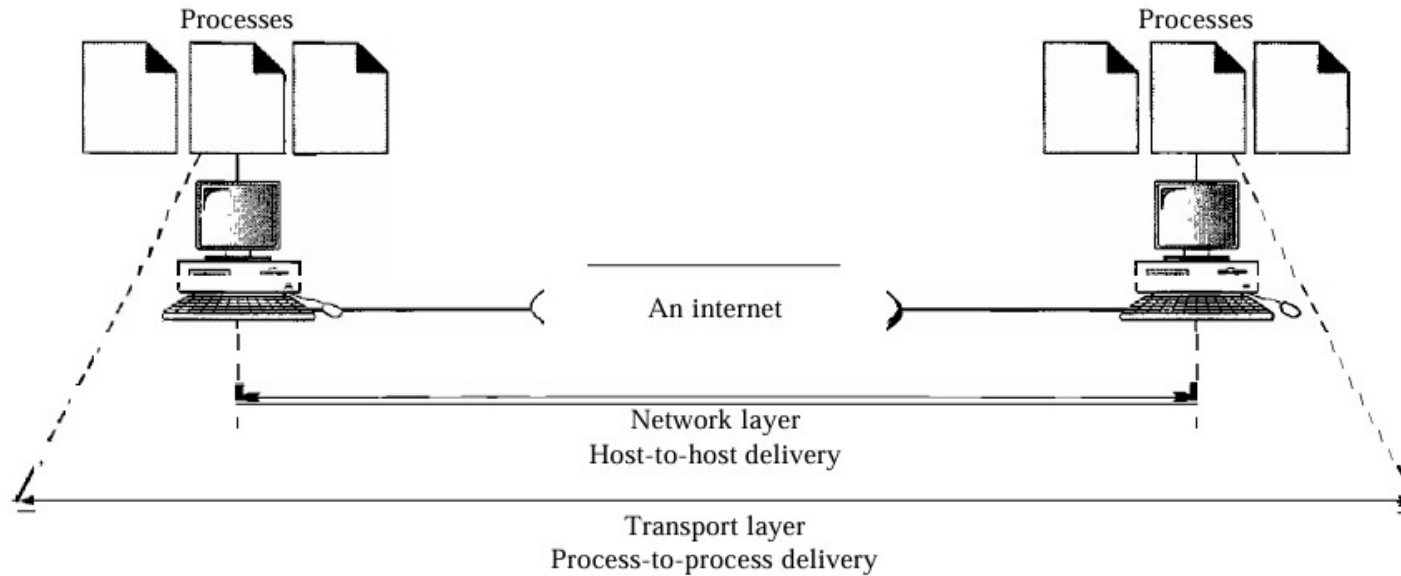
The transport layer is responsible for process-to-process delivery of the entire message



SEGMENT 1 : [X9A#K2@]
SEGMENT 2 : [P\$7L!M8]
SEGMENT 3 : [Q]

Transport Layer

Source to destination delivery by the network layer.

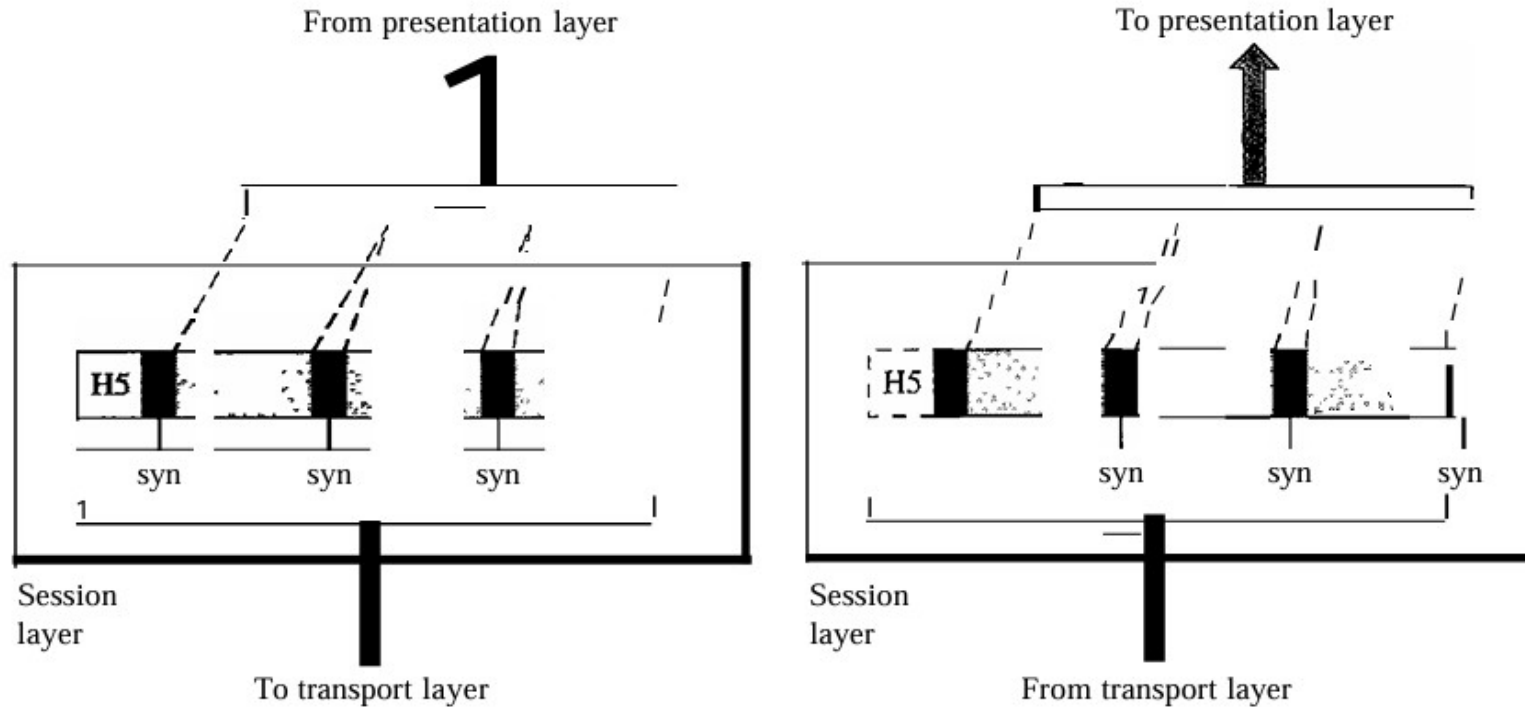


Functions of Transport Layer

- Service-point addressing
- Segmentation and reassembly
- Connection control
- Flow Control
- Error COntrol

Session Layer

The session layer ensures that two communicating devices establish, manage, and properly terminate a structured communication session for organized data exchange.



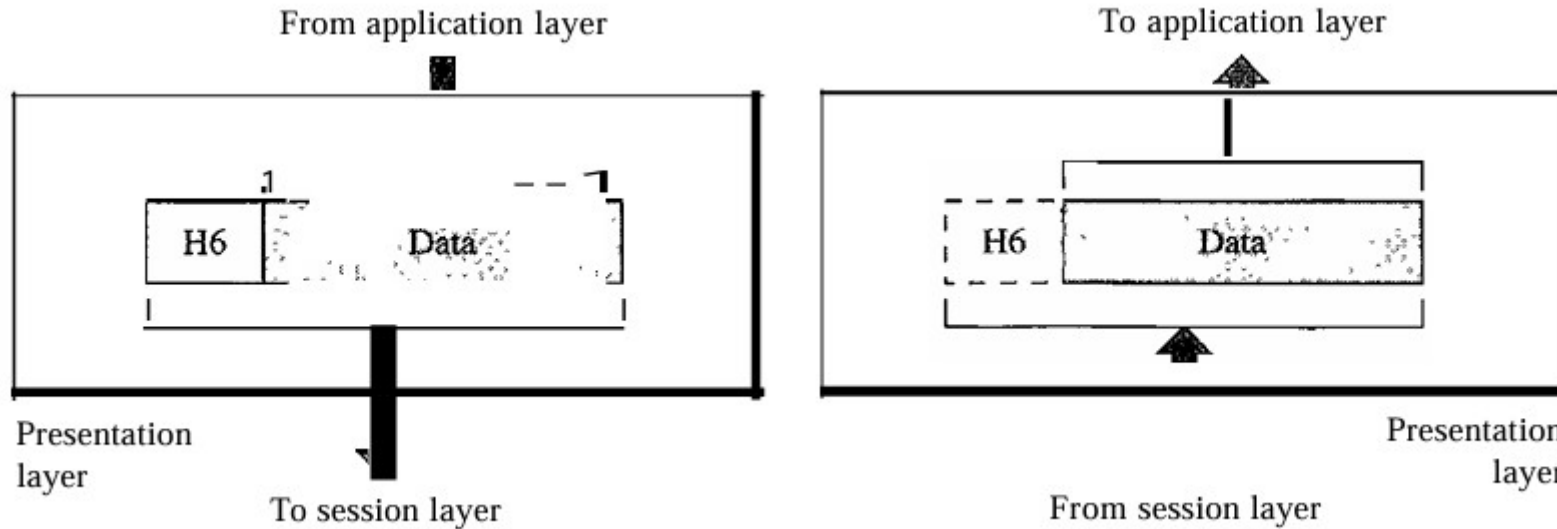
THIS IS A LARGE DATA MESSAGE

Functions of Transport Layer

- Dialog control. The session layer allows two systems to enter into a dialog. It allows the communication between two processes to take place in either half duplex (one way at a time) or full-duplex (two ways at a time) mode.
- Synchronization. The session layer allows a process to add checkpoints, or synchronization points, to a stream of data.

Presentation Layer

The presentation layer is concerned with the syntax and semantics of the information exchanged between two systems.



*PLAINTEXT:
THIS IS A LARGE DATA MESSAGE*

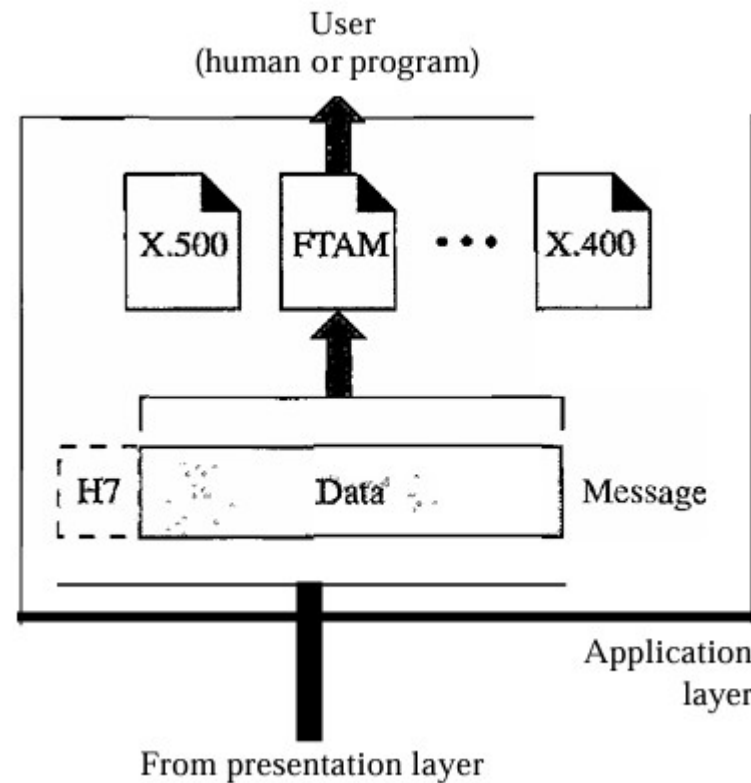
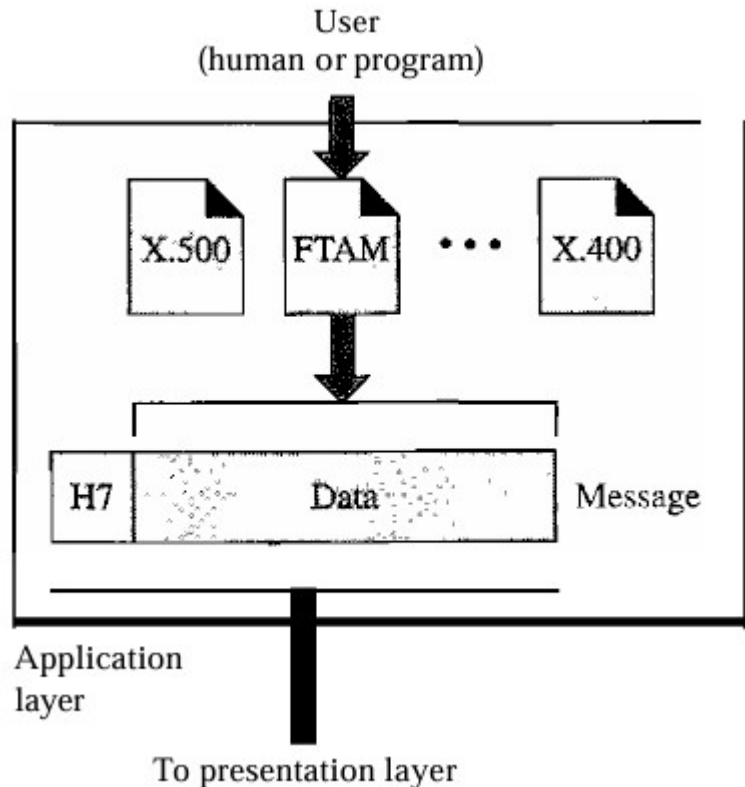
*ENCRYPTED DATA:
X9A#K2@P\$7L!M8Q*

Functions of Presentation Layer

- Translation. The processes(running programs)in two systems are usually exchanging information in the form of character strings, numbers, and so on. The information must be changed to bit streams before being transmitted.
- Encryption. To carry sensitive information, a system must be able to ensure privacy.
- Compression. Data compression reduces the number of bits contained in the information.

Application Layer

The application layer enables the user, whether human or software, to access the network.

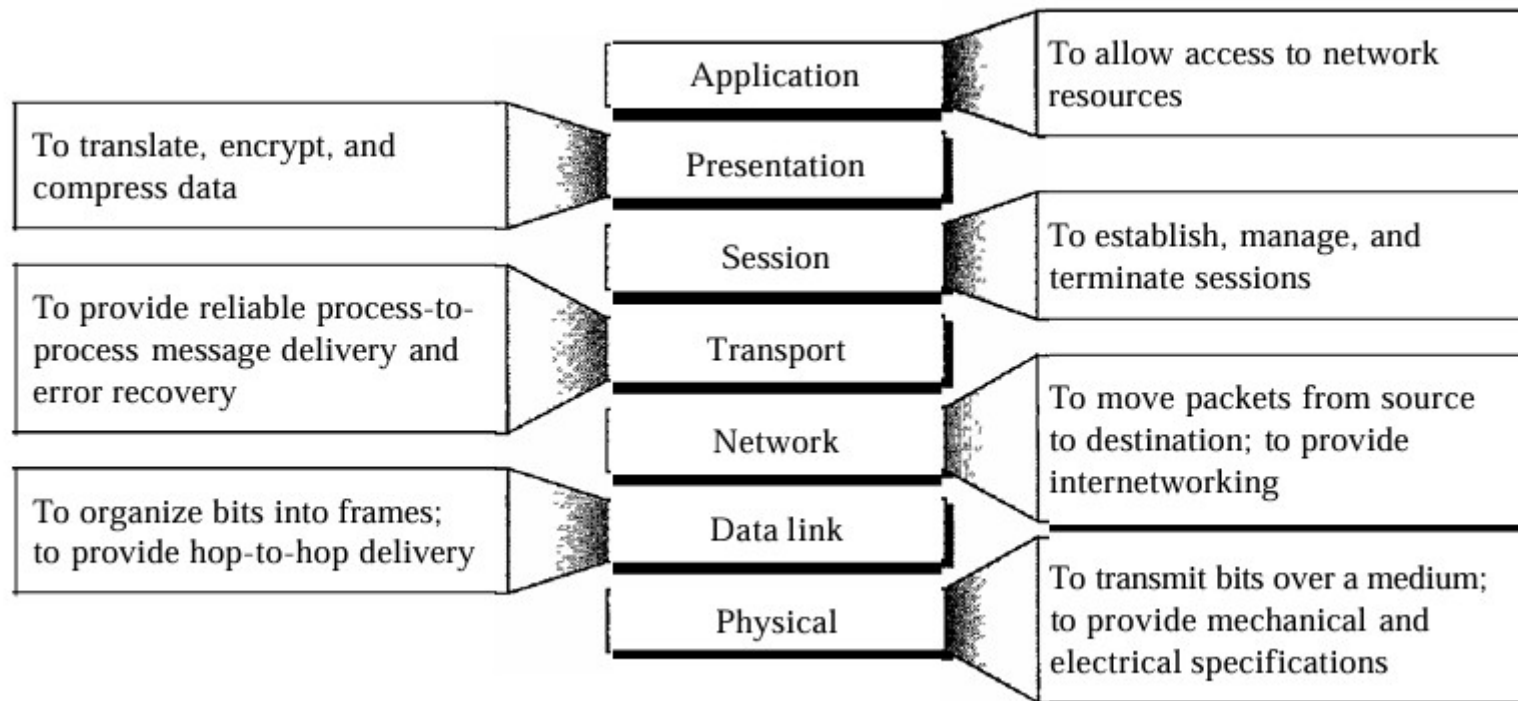


THIS IS A LARGE DATA MESSAGE

Functions of Application Layer

- File transfer, access, and management.
- Mail services.
- Directory services.

Summary



Question

Suppose a file of size 100 KB (kilobytes) is transmitted from a source host to a destination host through an intermediate network system. The network follows the OSI reference model, and each OSI layer introduces a processing delay of 1 *ms*. The source host is connected to the intermediate system by a *coaxial cable* of length 1000 km, and the intermediate system is connected to the destination host by another coaxial cable of length 1500 km. The signal propagation speed in the cable is 200,000 km/s, and the data transmission rate supported by the cable is 1000 kbps. Assuming store-and-forward transmission, determine the total time required for the complete file to reach the destination host.